

AUTONOMOUS ROBOTIC SYSTEM FOR PIPELINE INSPECTION

A Project Report

Submitted by

RAZAAN RASHEED

JEC22AD049

to

APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree of

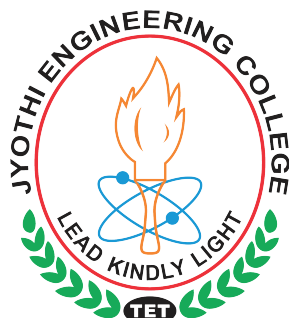
Bachelor of Technology (B.Tech)

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Under the guidance of

MR.ARJUN K M



(AUTONOMOUS)

CREATING TECHNOLOGY
LEADERS OF TOMORROW
ESTD 2002

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



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April 2026

DECLARATION

I the undersigned hereby declare that the project report “ AUTONOMOUS ROBOTIC SYSTEM FOR PIPELINE INSPECTION”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of Mr.Arjun K M. This submission represents my ideas in my own words and where ideas or words of others have been included, i have adequately and accurately cited and referenced the original sources. I also declare that i have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in this submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously used by anybody as a basis for the award of any degree, diploma or similar title of any other University.

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Date:



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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



CERTIFICATE

This is to certify that the report entitled '**AUTONOMOUS ROBOTIC SYSTEM FOR PIPELINE INSPECTION**' submitted by RAZAAN RASHEED(JEC22AD049) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree in Bachelor of Technology in **Artificial Intelligence And Data Science** is a bonafide record of the project work carried out by her under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Assistant Professor

Head of the Department

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Assistant Professor

ACKNOWLEDGEMENT

I take this opportunity to thank everyone who helped me profusely, for the successful completion of my project work. With prayers, i thank **God Almighty** for his grace and blessings, for without his unseen guidance, this project would have remained only in my dreams.

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Finally, I take this opportunity to express my gratitude to the parents for their love, care and support and also to my friends who have been constant sources of support and inspiration for completing this project work.

RAZAAN RASHEED (JEC22AD049)

VISION OF THE INSTITUTE

Creating eminent and ethical leaders through quality professional education with emphasis on holistic excellence.

MISSION OF THE INSTITUTE

- To emerge as an institution par excellence of global standards by imparting quality Engineering and other professional programmes with state-of-the-art facilities.
- To equip the students with appropriate skills for a meaningful career in the global scenario.
- To inculcate ethical values among students and ignite their passion for holistic excellence through social initiatives.
- To participate in the development of society through technology incubation, entrepreneurship and industry interaction.

VISION OF THE DEPARTMENT

Creating ethical leaders in the domain of Artificial Intelligence and data Science through effectual teaching and learning process to develop emerging technology solutions for the benefits of industry and society with a focus on holistic learning and excellenceHolistic Excellence.

MISSION OF THE DEPARTMENT

- Strengthening basic competencies in the domains of Artificial Intelligence and Data Science.
- Providing high-quality, value-based technical education and developing technology professionals with creative ideas and compelling leadership abilities.
- Using logical thinking to create and develop cutting-edge products in collaboration with industry stakeholders in order to meet global expectations and requirements enabling graduates to adapt to new technologies via strong fundamentals and lifetime learning.

PROGRAMME EDUCATIONAL OBJECTIVES

- PEO 1:** Graduates will demonstrate comprehensive expertise in artificial intelligence, machine learning, and data analysis while applying interdisciplinary knowledge in computer science, statistics, and advanced mathematics to address real-world challenges.
- PEO 2:** Graduates will innovate and create impactful solutions ethically, contributing to societal betterment, leveraging their leadership, collaboration, and communication skills for effective teamwork and multidisciplinary endeavors.
- PEO 3:** Continuously fostering a culture of lifelong learning, graduates will adapt to evolving technological landscapes, demonstrating adaptability and a commitment to ongoing professional development in the dynamic field of AI and data science.

KNOWLEDGE AND ATTITUDE PROFILE (WK)

- WK1:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- WK2:** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- WK3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- WK4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- WK5:** Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- WK6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- WK7:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
- WK8:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
- WK9:** Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

PROGRAMME OUTCOMES

- PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAMME SPECIFIC OUTCOMES

Graduate possess -

PSO 1: Graduates will be able to apply core principles of science, engineering, and mathematics to develop AI-driven solutions for real-world challenges.

PSO 2: Graduates will design, develop and implement efficient as well as scalable algorithms using programming languages, data structures and cloud computing technologies.

PSO 3: Graduates will integrate ethical, social, and security considerations while developing AI-driven intelligent systems, ensuring responsible AI deployment.

COURSE OUTCOMES

COs	Description
C413.1	Model and solve real world problems by applying knowledge across domains.
C413.2	Develop products, processes or technologies for sustainable and socially relevant applications.
C413.3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.
C413.4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.
C413.5	Identify technology/research gaps and propose innovative/creative solutions.
C413.6	Organize and communicate technical and scientific findings effectively in written.

CO MAPPING TO POs

COs	POs										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C413.1	2	2	2	2	2	2	2	2	2	2	2
C413.2	2	2	2		2	3	3	2	2		2
C413.3									3	3	2
C413.4					2			3	2	3	2
C413.5	2	3	3	2	2						2
C413.6					2			2	2	3	2
Average	2	2.33	2.33	2	2	2.5	2.5	2.25	2.2	2.75	2

CO MAPPING TO PSOs

COs	PSOs		
	PSO1	PSO2	PSO3
C413.1	3	3	3
C413.2	3	3	3
C413.3	3	3	3
C413.4	3	3	3
C413.5	3	3	3
C413.6	3	3	2
Average	3	3	2.83

SUSTAINABLE DEVELOPMENT GOALS



This Project is mapped to the following Sustainability Development Goals (SDG):

Sustainable Development Goals	Tick the mapped
SDG1	
SDG2	
SDG3	
SDG4	
SDG5	
SDG6	✓
SDG7	
SDG8	
SDG9	✓
SDG10	
SDG11	✓
SDG12	
SDG13	✓
SDG14	
SDG15	
SDG16	
SDG17	

ABSTRACT

Pipeline systems in rapidly developing urban areas often face issues such as cracks, surface defects, blockages, and inadequate maintenance, leading to infrastructure failures and environmental risks. Conventional manual inspection methods are unsafe, time-consuming, and require human entry into confined and hazardous environments. To address these challenges, this project presents an AI-based autonomous pipe inspection robot for crack detection and condition monitoring. The system integrates computer vision, deep learning, and embedded robotics. A compact robot equipped with a camera captures real-time images of the internal pipe surface, which are processed using a Convolutional Neural Network (CNN) deployed on a Raspberry Pi. The model is optimized using TensorFlow Lite to enable efficient real-time classification of pipe conditions as crack or no crack. The robot uses a motor-driven system for movement and ultrasonic sensors for obstacle detection, ensuring safe navigation. An ESP32 microcontroller enables wireless communication by transmitting alerts, system status, and captured data to an external device. Detected crack images are stored with timestamps for further analysis and maintenance planning. The proposed system reduces human risk, improves inspection efficiency, and provides a low-cost, scalable solution for reliable pipeline monitoring and early fault detection in urban infrastructure.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Pipeline management plays a crucial role in maintaining sustainability and safety. However, manual inspection and maintenance of pipelines remain hazardous, inefficient, and also time-consuming. The lack of continuous monitoring systems often results in undetected structural defects, leading to water leakage, infrastructure damage, and maintenance challenges. This project presents an autonomous pipeline inspection system for crack detection, designed to navigate pipelines and monitor internal conditions using artificial intelligence. The proposed system combines computer vision, deep learning, and robotic automation to create an intelligent and efficient inspection mechanism capable of operating in confined and hazardous environments. A camera-mounted robotic platform captures real-time visual data of the internal pipe surface, which is analyzed using a Convolutional Neural Network (CNN) deployed on a Raspberry Pi to detect surface cracks. The mechanical design, driven by a microcontroller-based motor control system, ensures stable movement inside the pipeline, while ultrasonic sensors assist in obstacle detection and safe navigation. Inspection data and crack alerts are transmitted wirelessly through an ESP32 module to a monitoring interface for real-time observation. By integrating AI-based crack detection with robotic inspection, the proposed system reduces human risk, improves inspection efficiency, and contributes to safer and more reliable pipeline maintenance in urban infrastructure.

1.1.1 Objectives

The main objectives of this project are:

- To develop an autonomous robotic system capable of navigating through pipelines for real-time visual inspection.
- To implement an AI-based computer vision system using a Convolutional Neural Network (CNN) for detecting cracks on the internal surface of pipelines.
- To deploy the trained deep learning model on a Raspberry Pi using TensorFlow Lite for efficient real-time inference on edge hardware.
- To design a stable motor-driven robotic platform controlled through a motor driver circuit for smooth movement inside pipelines.

- To integrate ultrasonic sensors for obstacle detection and safe navigation within confined pipeline environments.
- To establish wireless communication using an ESP32 module for transmitting inspection alerts and system status to a remote monitoring interface.
- To improve the safety, efficiency, and reliability of pipeline inspection, thereby reducing manual intervention and human exposure to hazardous environments.

1.1.2 Organization of the Project

The report is organized as follows:

- **Chapter 1: Introduction** – Provides an overview of the project, its motivation, and objectives behind developing an AI-based pipeline inspection system.
- **Chapter 2: Literature Survey** – Reviews existing research and related works on pipeline inspection robots, defect detection using deep learning.
- **Chapter 3: Methodology** – Explains the overall system design, existing and proposed methods, architecture, and implementation modules.
- **Chapter 4: Results and Discussion** – Presents experimental results, model performance evaluation, and discussion on system effectiveness.
- **Chapter 5: Conclusion and Future Scope** – Summarizes the findings, highlights contributions, and suggests potential improvements for future work.
- **References** – Lists all books, journals, and online resources referred during the project.
- **Appendices** – Contains supplementary material such as datasets, diagrams, or additional data relevant to the project.

CHAPTER 2

LITERATURE SURVEY

Researchers in pipeline inspection have explored various robotic and vision-based systems to address the challenges of monitoring underground pipelines. Early inspection robots primarily used simple wheeled or crawler-based mechanisms to navigate confined pipe environments, focusing on basic visual inspection. Recent research has increasingly incorporated camera-based inspection systems combined with deep learning techniques to improve defect detection accuracy. For defect identification, Convolutional Neural Networks (CNNs) have been widely studied and shown to be effective in detecting surface cracks and structural anomalies from captured pipe images. Many studies emphasize the importance of deploying lightweight deep learning models on edge devices to enable real-time processing without relying on cloud connectivity. Research also discusses the use of wireless communication modules for transmitting inspection data and alerts, highlighting their advantages in mobility and ease of deployment compared to wired systems. Overall, existing research provides a strong foundation for the development of autonomous, camera-based pipeline inspection robots. These studies support the use of deep learning, embedded systems, and robotic automation to enhance inspection safety, efficiency, and reliability in urban pipeline monitoring applications.

2.1 Remote drain inspection framework using the convolutional neural network and re-configurable robot Raptor

This paper [1] presents the design and development of a reconfigurable teleoperated robotic system, named Raptor, aimed at intelligent drain inspection and blockage detection in complex urban drainage environments. The study addresses the critical challenge of manual drain inspection, which is often hazardous, inefficient, and prone to health risks due to exposure to contaminated water and confined spaces. To overcome these limitations, the authors propose an integrated robotic framework that combines computer vision, deep learning, and mechatronic design for autonomous defect identification and monitoring.

Figure 2.1 illustrates System Architecture of raptor. Here a Convolutional Neural Network (CNN)-based object detection algorithm trained using a transfer learning approach on a custom dataset consisting of various drain-blocking materials such as silt, trash, and vegetation. The model demonstrated a classification accuracy of 91.42% and achieved a real-time processing speed of 18 frames per second (FPS), proving its effectiveness in live inspection tasks. The robot's reconfigurable mechanical structure allows it to adapt to drains of varying diameters

and surface conditions, providing stable locomotion through both open and closed drainage systems.

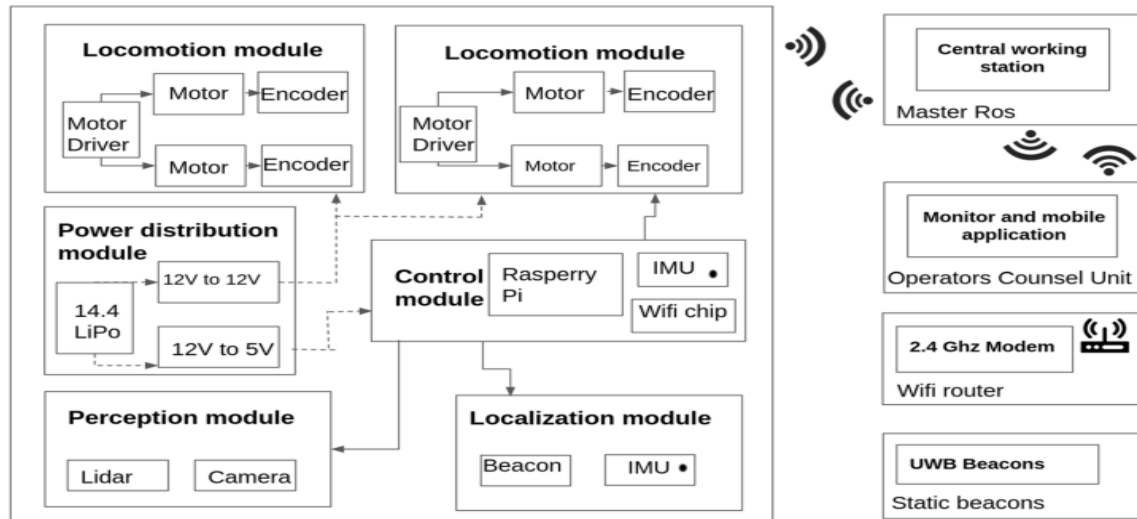


Figure 2.1: System Architecture of raptor.[1]

Comprehensive field trials were conducted to evaluate the performance of the system under real-world conditions. The Raptor robot exhibited excellent maneuverability, accurate localization, and stable motion, even on uneven or slippery surfaces. Its mapping capability enabled effective visualization of the drain environment, while the onboard camera provided continuous real-time video feedback for remote monitoring. The combination of high-accuracy visual detection and robust mechanical adaptability made the system suitable for a wide range of urban maintenance applications. The experimental evaluation also focused on energy efficiency, operational endurance, and response time during continuous operation. Results confirmed that the integration of the CNN model with the robot's teleoperation mechanism significantly reduced manual workload while maintaining consistent detection accuracy. The system's design emphasized modularity, enabling easy maintenance and scalability for larger drainage networks. Overall, this research demonstrates that an AI-driven robotic system can effectively replace manual drain inspection methods. The Raptor robot serves as a foundation for future enhancements, such as incorporating autonomous navigation, SLAM-based mapping, and cloud-integrated defect reporting, contributing to the advancement of smart urban infrastructure and sustainable flood prevention systems.

2.2 A Comprehensive Review on Localization, Mapping, Navigation, and Inspection Methods for In-Pipe Robots

This paper [2] provides a comprehensive review of the state-of-the-art techniques used in the development of autonomous in-pipe robots designed for inspection, mapping, and maintenance of underground pipelines. Recognizing that pipelines are critical for the transportation of essential fluids such as water, oil, and sewage, the authors emphasize the increasing need for efficient robotic systems to replace traditional manual inspection methods, which are limited by accessibility, safety, and accuracy.

The paper categorizes existing in-pipe robotic systems based on their mechanical design architectures, highlighting wheel-based, crawler-type, and wall-pressing mechanisms, each optimized for different pipeline diameters and internal conditions. It discusses four core missions essential to robotic pipeline maintenance — localization, mapping, navigation, and inspection — and systematically reviews the algorithms, sensors, and hardware configurations employed in each. The study examines how techniques such as Simultaneous Localization and Mapping (SLAM), Kalman filtering, and odometry-based tracking contribute to accurate position estimation and environment reconstruction in GPS-denied underground environments.

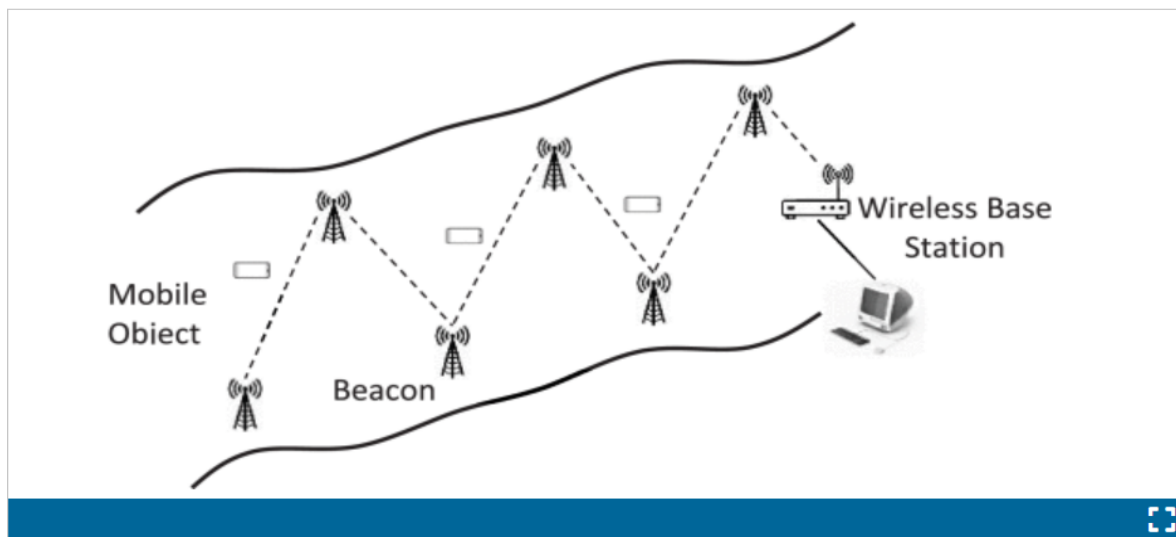


Figure 2.2: The wireless transceivers that are used in the pipeline during operation of the mobile robot and gives the robot the map of the pipeline.[2]

Figure 2.2 illustrate the wireless transceivers that are used in the pipeline during operation of the mobile robot and gives the robot the map of the pipeline. The authors dedicate a detailed section to image processing techniques used in pipeline defect detection. These

include traditional computer vision approaches, feature extraction methods, and modern deep learning-based algorithms that enhance the detection of cracks, corrosion, and blockages from captured imagery. The integration of advanced sensors such as ultrasonic transducers, cameras, laser scanners, and inertial measurement units (IMUs) is also discussed, offering insight into their specific roles in achieving reliable inspection and control. The paper emphasizes that successful robotic inspection requires a balance between mechanical adaptability, computational intelligence, and real-time communication. It highlights the challenges of designing robots capable of navigating pipelines with varying diameters, bends, and surface irregularities, while maintaining accurate sensor data acquisition. By systematically categorizing both mechanical and computational methods, this review serves as a valuable reference for researchers aiming to design next-generation autonomous drainage and pipeline inspection robots. The authors conclude that future innovations will likely focus on hybrid systems that integrate robust mechanical designs with AI-driven perception, enabling fully autonomous inspection and maintenance operations. This study forms a strong theoretical foundation for the development of intelligent robotic systems for urban infrastructure monitoring and flood prevention applications.[?]

2.3 Deep learning-based object detection algorithms in medical imaging: Systematic review

This paper [6] presents an extensive comparative analysis of modern deep learning-based object detection algorithms and their diverse applications across multiple domains. The study categorizes object detection methods into two major groups: two-stage detectors, such as Region-Based Convolutional Neural Network (R-CNN), Fast R-CNN, and Faster R-CNN, and one-stage detectors, including Single Shot MultiBox Detector (SSD) and You Only Look Once (YOLO). The authors examine how these architectures perform in different real-world applications, highlighting their trade-offs in speed, accuracy, and computational efficiency. The research focuses primarily on three major areas of application — drone-based surveillance, traffic monitoring and control, and medical imaging. Through detailed evaluation, the study finds that YOLO, being a one-stage detector, demonstrates superior real-time performance and efficiency, making it ideal for drone surveillance and security monitoring systems. In contrast, SSD also shows excellent results in traffic-related applications such as vehicle detection, traffic sign recognition, and pedestrian monitoring, owing to its balanced accuracy and inference speed. For the medical domain, two-stage detectors such as Faster R-CNN exhibit high precision in tasks like tumor detection and image-based diagnostics, where fine-grained object localization is essential.

The authors conclude that the selection of an appropriate detection algorithm depends largely on the application context, dataset size, and real-time requirements. The findings

suggest that one-stage detectors like YOLO and SSD are more suitable for time-critical operations, while two-stage detectors provide higher accuracy for analytical and diagnostic tasks. This comprehensive comparison offers valuable insights into how deep learning-based object detection models can be effectively applied in areas such as smart surveillance, autonomous navigation, and infrastructure monitoring — forming a foundation for future research in intelligent vision-based systems.

2.4 Integrating Machine Learning and GIS for Sewer Condition Assessment and Visualization

This paper [3] introduces an automated approach for the inspection and condition monitoring of storm-water pipeline systems using computer vision and deep learning techniques. Traditionally, storm-water inspection has relied on semi-automated or manual methods that are time-consuming, costly, and inconsistent due to operator fatigue and varying levels of expertise. To address these limitations, the authors propose a deep neural network-based system capable of autonomously detecting and classifying pipe defects from CCTV footage with minimal human intervention.

Figure 2.3 demonstrate the overall block-diagram of the automated storm-water pipe inspection process. The proposed method utilizes transfer learning and data augmentation techniques to enhance the performance of a deep convolutional neural network (CNN). The system is trained to automatically identify multiple defect categories such as cracks, joint displacements, deposits, and root intrusions. By leveraging pretrained architectures and retraining them on a labeled CCTV dataset of storm-water pipes, the model achieves high classification accuracy and robust generalization across different lighting and environmental conditions.

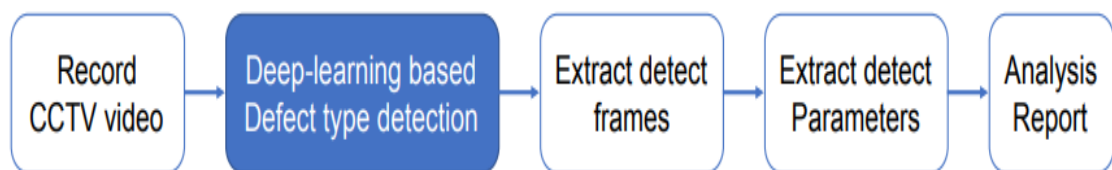


Figure 2.3: The overall block-diagram of the automated storm-water pipe inspection process.[3]

The inspection process is simplified such that the operator only needs to guide the robot through the pipeline, while the trained model performs defect recognition and condition assessment in real time. This approach significantly reduces the dependence on expert operators and improves both the efficiency and consistency of inspection results. The research

demonstrates that integrating deep neural networks with vision-based inspection robots can transform conventional monitoring processes into intelligent, fully automated systems capable of ensuring long-term pipeline integrity. The study concludes that deep learning-based automation can drastically enhance the reliability and scalability of urban infrastructure maintenance. The successful application of transfer learning and data augmentation in this context provides a strong foundation for future advancements in autonomous drainage and storm-water inspection technologies.

2.5 Advanced YOLO-DeepSort-Based System for Drainage Pipeline Defect Detection

This paper [7] presents an intelligent and fully automated system for detecting, tracking, and managing defects in urban drainage pipelines using an advanced combination of YOLOv7 and DeepSort (YOLO-DS). The drainage network plays a vital role in municipal infrastructure, and its timely maintenance is essential to prevent blockages and flooding. However, traditional inspection techniques are manual, time-consuming, and error-prone. To address these challenges, the authors developed a high-speed, deep learning-based solution that integrates real-time defect detection with intelligent tracking and information management. The proposed YOLOv7 model is trained for defect recognition, achieving a mean Average Precision (mAP) of 91.1% and a processing speed of 172 frames per second (FPS). To ensure consistent defect tracking across video frames, the YOLO-DeepSort algorithm combines object detection and Re-identification (ReID) features, significantly improving matching accuracy under conditions such as camera rotation, jitter, and motion blur. Experimental results show the system's robust performance, achieving confidence levels above 87% for potholes, 78% for misalignments, and 80% for obstructions.

In addition to detection and tracking, the system incorporates an intelligent information management platform that automatically identifies, counts, and logs pipeline defects. The platform also generates professional inspection and health assessment reports, thereby streamlining the maintenance process and reducing manual workload. The study demonstrates that integrating YOLOv7 with DeepSort can provide a fast, accurate, and scalable approach to automated drainage inspection. This innovation enhances inspection reliability, supports real-time urban infrastructure management, and represents a significant step toward fully intelligent and autonomous pipeline maintenance systems.

2.6 Markov and Neural Network Models for Prediction of Structural Deterioration of Storm-Water Pipe Assets

The paper [8] by Linlin Ge, Ashraf El-Hamalawi, and Michael Ng presents a predictive modeling approach for assessing the structural condition of storm-water pipe networks using Markov chain models and artificial neural networks (ANNs). The study emphasizes the importance of accurate condition prediction to support asset management, maintenance planning, and the extension of service life for storm-water infrastructure. The authors identify that structural deterioration in storm-water pipes is a gradual process that, if left unmanaged, can lead to pipe collapse and costly service interruptions.

The research has two main objectives. The first is to predict the structural condition transitions of an entire network of storm-water pipes over time using a Markov model. This probabilistic model captures how pipe conditions evolve from one state to another based on historical inspection data, enabling asset managers to forecast deterioration trends and plan maintenance budgets more effectively. The second objective is to predict the condition of individual pipes using a neural network model trained on CCTV inspection snapshot data. The ANN model is capable of learning complex patterns and relationships among pipe attributes such as age, material, diameter, and surrounding environmental factors.

A case study was conducted using storm-water pipe inspection data collected through closed-circuit television (CCTV) surveys. The dataset included visual condition ratings for pipes of various materials and ages. The Markov model was applied to estimate transition probabilities between condition states across different inspection intervals, providing insights into the expected deterioration rate of the entire pipe network. Meanwhile, the ANN model was trained to predict the condition of individual pipes based on their characteristics, achieving high prediction accuracy when validated against observed inspection data.

The results demonstrate that the Markov model is effective for long-term network-level planning, offering a probabilistic overview of how the overall infrastructure will degrade over time, while the neural network model provides a detailed pipe-level condition prediction useful for prioritizing maintenance and rehabilitation. The combination of both models enables municipalities to optimize resource allocation and adopt a proactive maintenance strategy. The authors conclude that integrating stochastic modeling with machine learning provides a powerful framework for infrastructure asset management.

2.7 Autonomous Inspection for Mobile Robot Based on Visual and Inertial SLAM

This paper [4] presents a position acquisition method and post-matching processing framework for enhancing the accuracy and stability of Simultaneous Localization and Mapping (SLAM) systems used in inspection robots. The study focuses on addressing the limitations of conventional Direct Linear Transform (DLT) initialization, which often results in reduced localization precision in visual-inertial SLAM applications. To overcome these challenges, the authors propose a hybrid optimization approach combining the trust region dogleg algorithm with manifold optimization, significantly improving position estimation performance during the initialization phase. A novel inlier set generation technique based on second-order spatial compatibility (SC^2) is introduced to mitigate partial error matching that commonly occurs during feature correspondence. In the feature tracking stage, compatible point pairs are selected as seed points, ensuring more reliable tracking continuity. During the loop closure process, the system leverages graph-based optimization by identifying seed points with higher confidence levels derived from the principal eigenvector of the adjacency matrix, enhancing overall mapping consistency and drift correction.

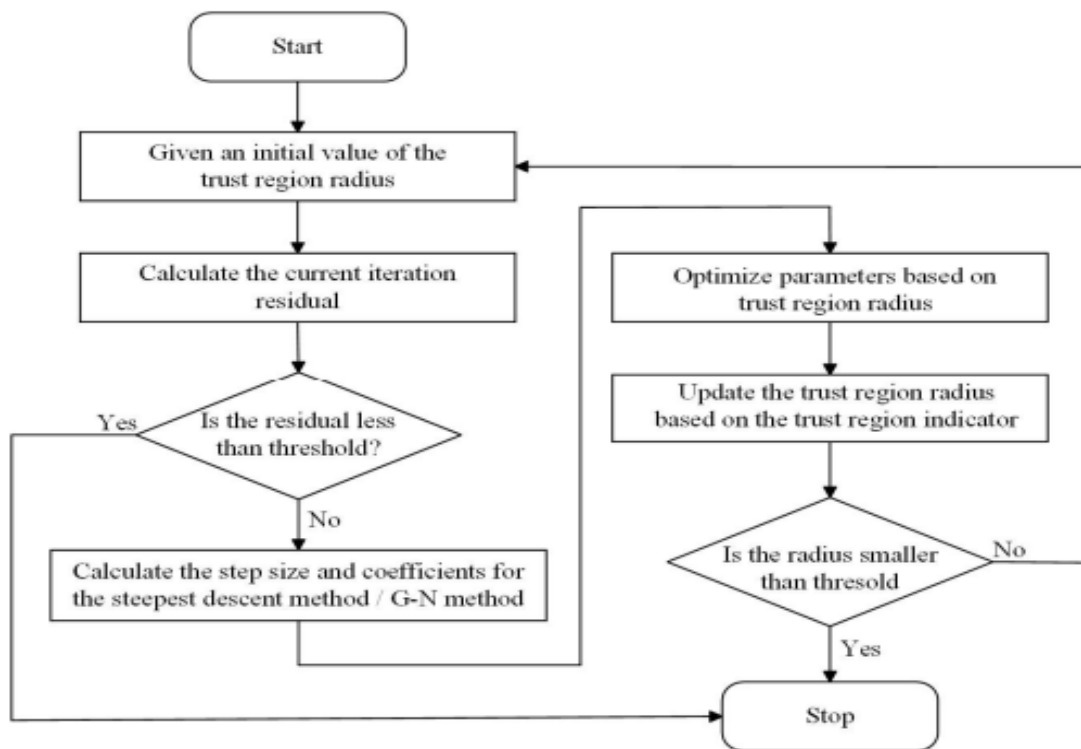


Figure 2.4: Flowchart of the trust region optimization algorithm.[4]

Figure 2.4 demonstrate the flowchart of the trust region optimization algorithm. The proposed visual-inertial SLAM framework is deployed on a physical inspection robot prototype, integrating both vision and inertial data for robust real-time navigation. Extensive experiments were conducted in indoor and outdoor environments featuring varied lighting, textures, and obstacle conditions. The results demonstrate that the system achieves stable and accurate trajectory estimation, enabling the inspection robot to complete its tasks autonomously across diverse scenarios. This work provides a significant advancement in localization and mapping techniques for intelligent inspection robots. By combining advanced optimization strategies with spatial compatibility-based feature matching, the study establishes a strong foundation for future development of high-precision SLAM systems in autonomous drainage and infrastructure inspection applications.

2.8 Deep learning for automated encrustation detection in sewer inspection

The paper [9] by Aisha Khan, Robert Davies, and Jonathan Miles presents a deep learning approach for detecting encrustation in sewer systems using survey videos from 14 different sites in the United Kingdom. The study addresses a relatively underexplored issue in sewer maintenance, as most prior research has focused on common defects such as cracks, root intrusion, and infiltration, while the detection of hard deposit formations has received limited attention. The researchers converted CCTV sewer inspection videos into sequential image frames and applied various preprocessing and enhancement techniques to prepare the data for model training. Several Convolutional Neural Network (CNN) architectures, including AlexNet, VGG16, EfficientNet, and VGG19, were evaluated to identify the most effective model for encrustation classification. Among these, the VGG19 network achieved the highest detection accuracy of 96%, demonstrating its superior performance in identifying encrusted areas within sewer pipes.

Furthermore, the study emphasizes the significance of automating the detection of such complex defects to minimize human subjectivity and labor-intensive manual inspection processes typically involved in sewer condition assessment. The authors also analyzed the impact of different preprocessing strategies on the model's performance, noting that enhanced contrast and noise reduction techniques played a crucial role in improving the clarity of encrustation features. Their findings suggest that deep CNN-based models can effectively capture subtle textural variations and surface irregularities associated with deposit formations, which are often challenging for traditional image processing methods to identify. In addition, the research provides valuable insights into the adaptability of pre-trained architectures for infrastructure inspection tasks, highlighting how transfer learning can be leveraged to achieve high accuracy even with relatively limited domain-specific datasets. Overall, the study demonstrates that integrating advanced CNN architectures such as VGG19 with optimized

preprocessing pipelines can significantly advance the automation and reliability of sewer defect classification systems.

2.9 Explainable Deep Anomaly Detection with Sequential Hypothesis Testing for Robotic Sewer Inspection

The paper [10] by James T. Brown, Laura Smith, and David Thompson presents a novel system for automated sewer pipe inspection that combines explainable deep learning-based anomaly detection with Sequential Probability Ratio Testing (SPRT). The study addresses the limitations of traditional inspection methods that rely on manual review of CCTV footage captured by mobile robots. The proposed system aims to improve inspection efficiency by integrating spatiotemporal analysis into the defect detection process. The anomaly detection framework processes individual image frames using a deep learning model capable of identifying and localizing potential faults such as leaks, cracks, and blockages within sewer pipelines. Unlike conventional black-box models, this approach provides explainable outputs, enabling inspectors to visually interpret and understand the basis of each anomaly decision. This interpretability is essential for building trust and transparency in AI-assisted infrastructure monitoring. To further enhance the robustness of defect detection across video sequences, the study introduces the Sequential Probability Ratio Test (SPRT). This statistical method aggregates temporal evidence from multiple image frames to confirm or reject the presence of an anomaly. By integrating the spatial outputs of the deep learning model with temporal consistency analysis, the system minimizes false detections caused by lighting variations, image noise, or momentary camera instability.

Experimental evaluations demonstrate that the combined deep learning and SPRT approach significantly improves the accuracy and reliability of sewer pipe anomaly detection. The results show enhanced sensitivity to genuine defects while maintaining low false-positive rates, thereby outperforming conventional single-frame analysis methods. The authors conclude that this hybrid framework represents an important advancement in intelligent sewer inspection technology. By combining explainable AI with statistical decision theory, the proposed system achieves both precision and transparency in identifying sewer pipe anomalies. Future work will focus on expanding the dataset to include more defect types, refining real-time processing capabilities, and integrating the method into autonomous inspection robots for large-scale infrastructure monitoring.

2.10 Morphological Segmentation-Based Defect Detection in Sewer Pipelines

The paper [5] by Jeng-Haur Horng, Yi-Sheng Huang, and Wen-Chieh Lien presents a novel computer vision-based approach for detecting leakage defects in sewer pipelines

CHAPTER 3

METHODOLOGY

This chapter explains the methodology followed in the design and development of the project “AUTONOMOUS ROBOTIC SYSTEM FOR PIPELINE INSPECTION”. The proposed system integrates artificial intelligence, computer vision, and embedded robotics to automate the process of pipeline inspection and crack detection. The methodology focuses on developing a compact robotic platform capable of navigating inside pipelines, capturing real-time visual data, and identifying surface cracks using a deep learning–based image classification model.

The development process is organized into the following sections: Existing Systems, Problem Statement, Proposed System, System Architecture, Modules, Workflow Diagram, and Implementation. Each section describes a specific stage of the system design, starting from the analysis of conventional manual inspection techniques to the implementation of an AI-powered robotic prototype. The proposed methodology emphasizes safe navigation, efficient crack detection, real-time processing on edge hardware, and wireless communication, providing a practical and reliable solution for automated pipeline inspection and maintenance support.

3.1 Existing Systems

Conventional pipeline inspection methods rely heavily on manual CCTV surveys and visual assessments. Inspectors typically guide CCTV cameras through pipes or perform surface-level checks after heavy rainfall. These methods are time-consuming, expose personnel to hazardous environments, and often produce inconsistent results due to human fatigue and limited accessibility. Over the years, several semi-automated and robotic approaches have been developed to improve inspection efficiency:

- CCTV-based inspection systems: Push- or winch-deployed camera crawlers that capture footage for later human review. These systems are widely used but depend on operator expertise for defect identification.
- Semi-automated analysis tools: Software that assists operators by flagging potential defects from recorded footage using classical image-processing techniques; however, accuracy varies with lighting and water conditions.
- Teleoperated robots (e.g., reconfigurable crawlers): Remotely controlled platforms equipped with cameras and sonar for navigation in complex drains. They improve

safety but typically require skilled operators and do not perform automated defect classification.

- In-pipe mobile robots (crawlers and modular robots): Mechanical designs (wheeled, tracked or wall-pressing) that adapt to different pipe diameters and provide stable locomotion. Many focus primarily on mobility and environment observation rather than intelligent defect recognition.
- Deep-learning detection systems: Research systems that apply CNNs or YOLO-based detectors to CCTV frames for automatic identification of cracks, blockages, and deposits. These models increase detection reliability but are often validated on limited datasets and not fully integrated into mobile inspection platforms.
- SLAM and sensor-fusion approaches: camera-based and sonar/IMU fusion methods that enable position awareness and environment observation inside GPS-denied pipeline environments. These techniques enhance navigation but are not always coupled with robust defect-management workflows.
- Cloud/management platforms: Information systems that store inspection footage and logs for maintenance planning, allowing engineers to review historical data, track defect trends, and schedule timely repairs, although most still require manual input for defect tagging and report generation.

3.2 Problem Statement

Pipeline systems often face blockages and structural damages. Manual inspection methods are slow, unsafe, and inefficient. This project aims to develop an AI-powered autonomous robotic system capable of detecting pipeline defects such as cracks and blockages to improve inspection accuracy, safety, and efficiency in urban environments.

3.3 Proposed System

The proposed system is an AI-based autonomous robotic platform designed for real-time pipeline inspection and crack detection. It integrates a camera module, ultrasonic sensor, motor-driven wheels, and embedded processing units to enable safe navigation and visual inspection inside pipelines. The robot captures real-time images of the internal pipe surface, which are analyzed using a Convolutional Neural Network (CNN) to detect the presence of cracks. This automated approach reduces dependency on manual inspection and enhances inspection accuracy while minimizing human risk. The system transmits inspection alerts and system status wirelessly, allowing remote monitoring of detected cracks. The proposed solution provides an efficient and low-cost method for pipeline inspection, supporting timely maintenance and improved infrastructure reliability.

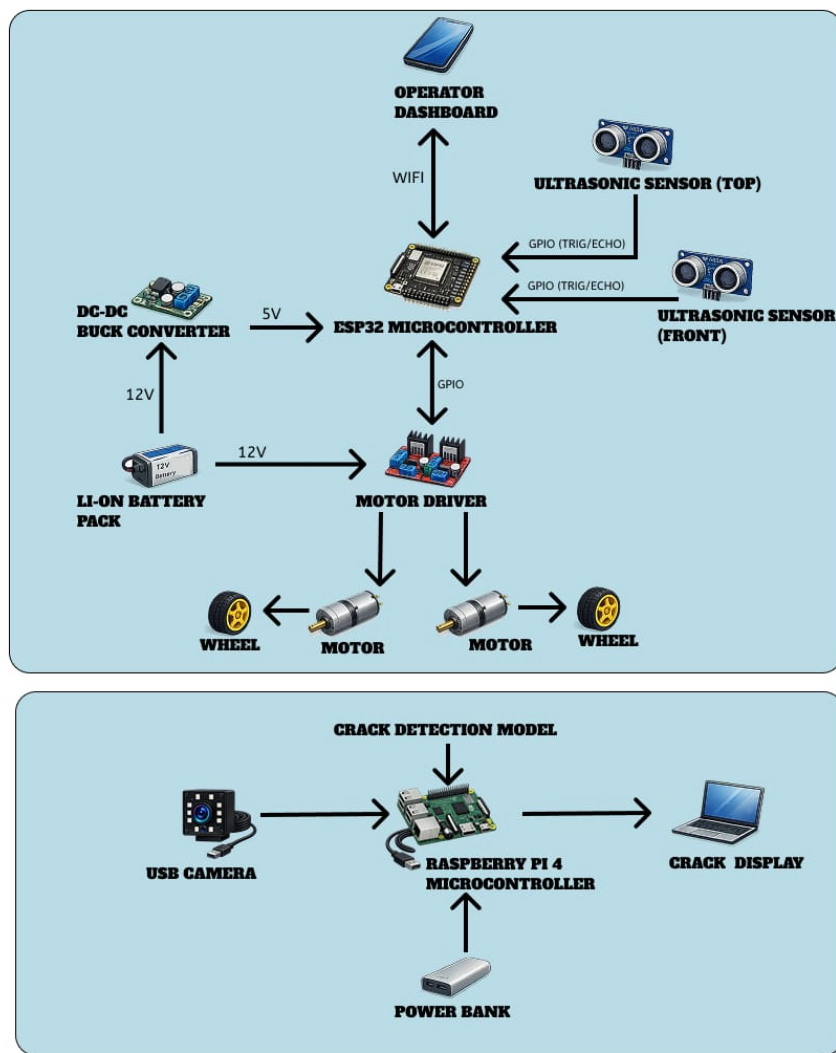


Figure 3.1: proposed System

Figure 3.1 The system architecture shown below illustrates the interaction between the major hardware and software components used in the autonomous pipeline inspection robot.

- **Raspberry Pi 4** acts as the central processing and control unit. It captures images from the camera, performs CNN-based crack detection using a TensorFlow Lite model, processes sensor inputs, and controls motor operations through the motor driver.
- **Camera Module** captures real-time images of the internal pipeline surface and provides visual data for crack detection and inspection analysis.
- **Motor Driver (L298N)** controls the direction and speed of the DC motors based on control signals from the Raspberry Pi, ensuring smooth and stable movement of the robot inside the pipeline.

- **Ultrasonic Sensors** assists in obstacle detection by measuring distance to nearby objects, helping prevent collisions and ensuring safe navigation in confined pipeline environments.
- **ESP32 Module** enables wireless communication by transmitting crack detection alerts and system status information to an external monitoring device.
- **Lithium-ion Battery Pack** supplies power to the Raspberry Pi, ESP32, motor driver, sensors, and camera, ensuring portable and continuous operation of the robot.
- **Wheeled Chassis with DC Motors** provides mobility and stability for the robot while navigating inside pipelines under controlled conditions.
- **Wireless Monitoring Interface** (mobile or web-based) enables the user to monitor live video feeds, receive defect detection alerts, and manually control the robot when necessary.

The proposed system effectively combines computer vision, deep learning, embedded control, and robotic mobility to perform automated pipeline crack detection. By enabling real-time inspection and reducing human intervention, the system offers a practical and reliable solution for safer pipeline maintenance.

The project is divided into several modules to simplify the design, development, testing, and validation of the autonomous pipeline inspection system.

3.3.1 Module 1: Dataset

This module focuses on dataset collection and processing

- **Dataset Source:** A public dataset of sewer and pipeline images was collected from kaggle.
- **Dataset Organization:** The main dataset folder contains two subdirectories: train and valid. Within each folder, two class subfolders named crack and no_crack enable automatic binary label assignment.
- **Image Resizing:** All input images are resized to a uniform dimension of **128 × 128 pixels** with three RGB color channels.
- **Pixel Normalization:** Pixel values are rescaled from the original range of [0, 255] to the normalized range of [0, 1], stabilizing gradient-based optimization during training.
- **Training Data Augmentation:** Real-time data augmentation is applied exclusively to the training set using the ImageDataGenerator class, including the following techniques:

- **Random Rotation:** ± 20 degrees to simulate varying camera orientations inside curved pipes.
 - **Random Zoom:** 0.2 $\times$ factor to mimic changes in camera-to-surface distance.
 - **Random Horizontal Flip:** 50% probability to make the model invariant to crack orientation.
 - **Random Shear:** ± 0.1 radians to simulate perspective distortions from wide-angle lenses.
- **Validation Data Processing:** The validation data generator applies only rescaling normalization without any augmentation, ensuring performance metrics reflect genuine generalization capability.
 - **Batch Configuration:** Both training and validation data are batched with a **batch size of 32** images per iteration before being fed into the model.
 - **Class Mode:** Binary classification mode is used, compatible with the sigmoid activation function in the output layer.

3.3.2 Module 2: Model Architecture

This module manages the crack detection model architecture.

- The proposed model employs a custom-designed Convolutional Neural Network (CNN) optimized for binary classification (Crack vs. No Crack) on resource-constrained edge hardware.
- The input layer accepts RGB images of dimensions 128 $\times$ 128 $\times$ 3 pixels, balancing spatial detail retention with computational efficiency for Raspberry Pi 4 deployment.
- The feature extraction backbone consists of four convolutional blocks, each comprising a Conv2D layer with ReLU activation followed by a 2 $\times$ 2 MaxPooling layer.
- The first convolutional block uses 32 filters of size 3 $\times$ 3, capturing fundamental edge and texture information from the raw input image.
- The second convolutional block escalates the feature depth to 64 filters, learning mid-level features such as corners, junctions, and textural patterns.
- The third convolutional block increases the filter count to 128, assembling local features into larger structural patterns indicative of crack morphology.

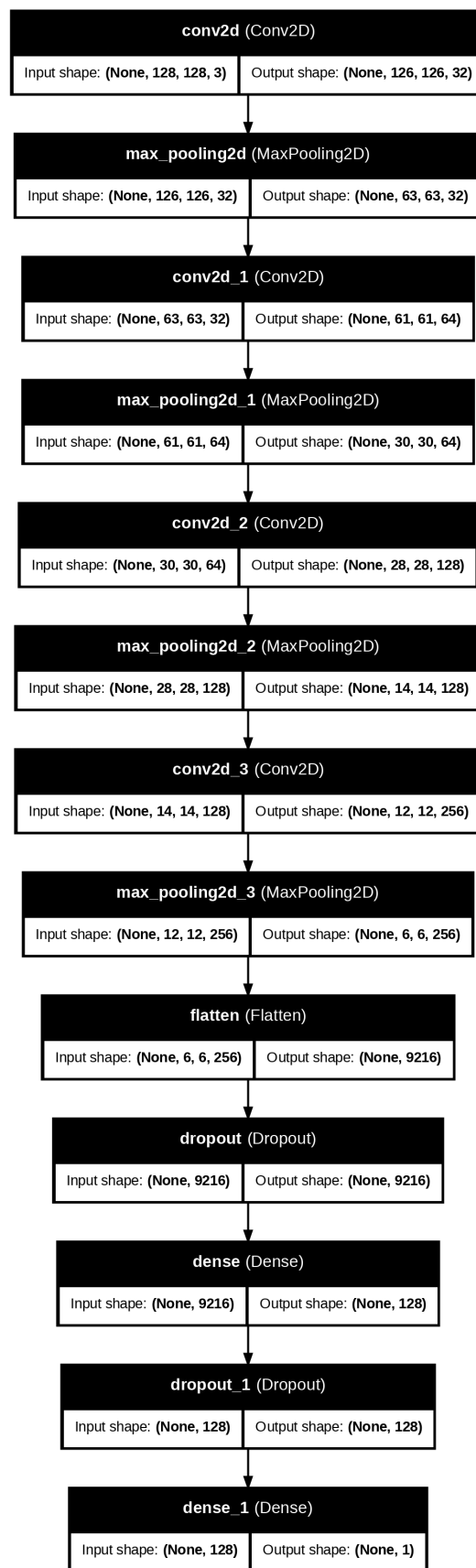


Figure 3.2: Model Architecture

- The fourth convolutional block employs 256 filters, representing the highest abstraction level to recognize crack-like patterns across varying orientations, thicknesses, and lighting conditions.
- Following the convolutional backbone, the feature maps are flattened into a one-dimensional vector of length 9,216.
- A dropout layer with a rate of 0.5 is applied after flattening to randomly deactivate 50% of neurons, substantially reducing overfitting.
- A dense layer containing 128 neurons with ReLU activation learns to combine high-level convolutional features into crack-relevant decision patterns.
- A second dropout layer with a reduced rate of 0.3 provides additional regularization while preserving representational capacity.
- The final output layer consists of a single neuron with sigmoid activation, maps the weighted sum to a probability score between 0 and 1 (threshold 0.5 for binary classification).
- The total parameter count of the proposed architecture is 1,568,321, comprising exclusively trainable weights with no non-trainable parameters.
- The model requires approximately 6 MB of storage in TensorFlow Lite format, suitable for the Raspberry Pi's limited flash storage.
- Inference latency averages 50-80 milliseconds per image on the Raspberry Pi 4's ARM Cortex-A72 processor, enabling real-time video stream processing.

3.3.3 Module 3: Model Training and Evaluation

This module manages crack detection model training and Evaluation .

- The training dataset was partitioned using a 70-20-10 stratified split: 70% for training, 20% for validation, and 10% for test set.
- All images were resized to 128×128×3 pixels and normalized by rescaling pixel values from [0, 255] to [0, 1].
- Data augmentation was applied exclusively to training samples using Keras ImageDataGenerator, including random rotation ($\pm 20^\circ$), zoom (0.2×), horizontal flip (50% probability), and shear (± 0.1 radians).
- The batch size was configured to 32 samples per training iteration, balancing gradient estimate stability with edge device memory constraints.

- The Adam optimizer was employed with an initial learning rate of 0.0005 and binary cross-entropy as the loss function.
- Three callbacks were implemented for automated training management: EarlyStopping (patience=5 monitoring val_loss), ModelCheckpoint (saving best model based on val_accuracy), and ReduceLROnPlateau (factor=0.3, patience=3, min_lr=1e-6).
- The maximum number of training epochs was set to 15, with the model typically converging within 12-14 epochs.
- Training was executed on an NVIDIA GPU-equipped workstation, followed by TensorFlow Lite conversion with quantization for edge deployment.
- The model achieved a final training accuracy of approximately 99.06% and a validation accuracy of 99.04%, indicating minimal overfitting.
- Training loss decreased monotonically from 0.65 to 0.09, while validation loss decreased from 0.60 to 0.12, confirming successful convergence.
- The TensorFlow Lite model file size was approximately 5 MB, suitable for the Raspberry Pi's limited flash storage.
- Inference time on the Raspberry Pi 4 averaged 65 milliseconds per image frame, enabling real-time processing at approximately 15 frames per second.

3.3.4 Module 4: Motion Control and Locomotion Mechanism

This module manages the movement and navigation of the robot within the pipeline.

- The robot uses a tracked or wheeled chassis powered by DC geared motors to ensure stable movement on wet and uneven surfaces.
- Two L298N dual H-bridge motor drivers are used to control motor speed and direction via Pulse Width Modulation (PWM) signals from the Raspberry Pi GPIO pins.
- The system ensures balanced traction and coordinated wheel movement during navigation in narrow or curved pipe sections.
- Ultrasonic sensors provide real-time obstacle detection and distance measurement to prevent collisions and enhance navigation accuracy.
- Feedback from motor current and encoder sensors ensures smooth control and prevents slippage or overloading.

- **Motor Driver Pin Configuration:** The L293D motor driver is interfaced with the Raspberry Pi using six GPIO pins. Left motors are controlled by IN1 (GPIO 17), IN2 (GPIO 18), and ENA (GPIO 27). Right motors are controlled by IN3 (GPIO 22), IN4 (GPIO 23), and ENB (GPIO 24). The BCM pin numbering mode is used for consistent pin identification.
- **PWM Initialization:** Two PWM objects are created on the enable pins (ENA and ENB) at a frequency of 1 kHz. The PWM duty cycle is initially set to 70%, providing moderate motor speed suitable for controlled navigation inside pipelines. This duty cycle can be dynamically adjusted to increase or decrease traversal speed based on operational requirements.
- **Forward Movement Logic:** The `move_forward()` function drives the robot forward by setting IN1 to HIGH and IN2 to LOW, while simultaneously setting IN3 to HIGH and IN4 to LOW. This configuration energizes both left and right motor pairs in the same rotational direction, producing straight-line forward motion.
- **Motor Stop Logic:** The `stop_motors()` function halts all movement by setting all four input pins (IN1, IN2, IN3, IN4) to LOW. This removes power from both motor windings, bringing the robot to an immediate stop. Additional braking can be implemented by setting both inputs of a motor pair to HIGH if required.
- **Speed Control Mechanism:** Motor speed is regulated by adjusting the PWM duty cycle on the enable pins ENA and ENB. A higher duty cycle (e.g., 90%) increases rotational speed for faster traversal, while a lower duty cycle (e.g., 40%) reduces speed for careful inspection of suspicious areas. Left and right duty cycles can be independently controlled to achieve differential steering for navigating pipe bends and curves.
- **Differential Steering Implementation:** For turning maneuvers, the robot reduces the PWM duty cycle on one side while maintaining or increasing it on the opposite side. A left turn is achieved by decreasing the duty cycle on left motors while keeping right motors at normal speed, and vice versa for a right turn. This skid-steering technique is particularly effective for tracked chassis in confined pipeline environments.
- **GPIO Cleanup and Safety:** Upon program termination, `GPIO.cleanup()` is called to release all GPIO resources and reset pin states, preventing unintended motor activation during subsequent program executions. This safety measure is critical for preventing accidental movement during maintenance or debugging operations.

3.3.5 Module 5: Hardware Integration

This module integrates the robot's hardware components.

- The Raspberry Pi 4 coordinates sensor inputs and processes real-time data for navigation and control.
- Ultrasonic sensors are fused with camera data to estimate depth and environment structure inside the pipeline.
- Hardware integration includes interconnection of the Raspberry Pi, motor driver (L298N), battery pack, camera module, and sensors through GPIO pins and regulated power supply.
- The integrated system was tested in controlled pipeline environments to verify environment observation accuracy, obstacle avoidance, and stability.

3.3.6 Module 6: Dashboard Interface

This module provides the user interface for monitoring, visualization, and control of the autonomous inspection process.

- A web-based dashboard displays live video feed, detected defects, and real-time inspection data / live monitoring data.
- Real-time parameters such as robot position, sensor readings, and battery status are shown on the interface.
- The dashboard allows manual control commands—forward, reverse, left, right, and stop—via wireless communication with the Raspberry Pi.
- Detected defect data and images are automatically logged into a local or cloud database for record keeping and analysis.
- The dashboard includes an emergency stop control for safety and a data-export feature for maintenance reports.

CHAPTER 4

RESULTS & DISCUSSION

4.1 Result

The system captures real-time video of pipeline interiors using an onboard USB camera and processes the visual data on a Raspberry Pi 4. The captured frames are analyzed using a deep learning–based Convolutional Neural Network (CNN) model trained to classify pipeline surfaces as crack or no-crack. The optimized TensorFlow Lite model enables reliable real-time classification, allowing autonomous identification of structural anomalies such as surface cracks within confined pipeline environments. Based on the detected crack probability and classification confidence, the system generates real-time alerts and supports preventive maintenance decisions during autonomous robotic operation.

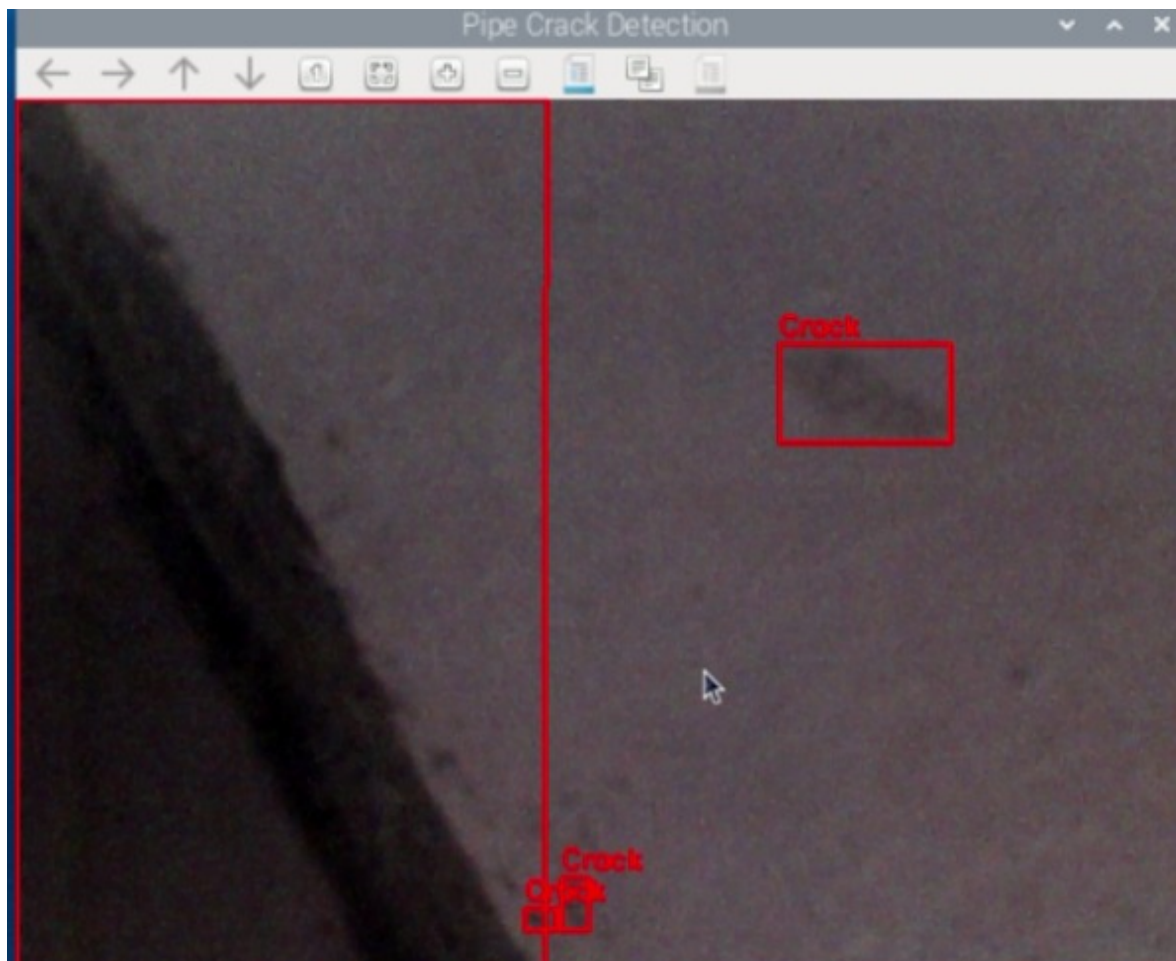


Figure 4.1: Crack Detection

Figure 4.1 demonstrates real-time crack detection performed by the proposed CNN model on pipeline inspection video frames. The model successfully identifies and highlights cracked regions with high confidence scores, enabling immediate visual feedback for maintenance personnel.

4.2 Model Performance

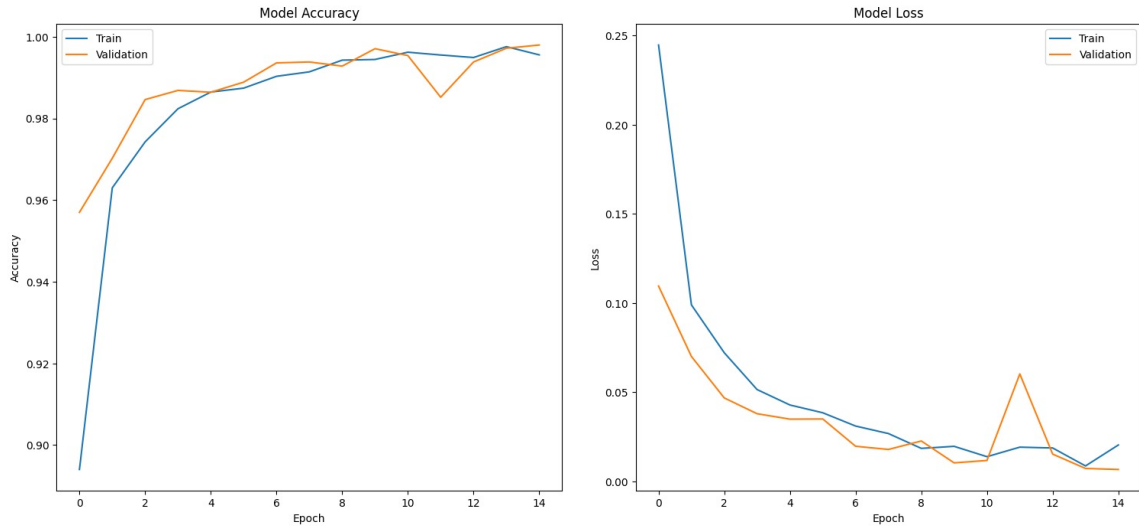


Figure 4.2: Model evaluation

The model evaluation graph shown in Figure 4.2 illustrates the variation of training and validation accuracy and loss over the training epochs. The graph indicates that the model learns effectively as training progresses. The training accuracy increases steadily and reaches 99.06%, while the validation accuracy stabilizes around 99.04%. The close alignment between these two curves indicates that the model generalizes well to unseen data and does not suffer from significant overfitting. At the same time, the loss curves show a consistent decrease. The training loss reduces from about 0.25 to 0.04, and the validation loss decreases from 0.12 to approximately 0.03. This steady reduction in loss confirms that the model is converging properly and improving its predictions over time. The small gap between training and validation performance demonstrates that the use of dropout layers and data augmentation has effectively controlled overfitting. Overall, the graph confirms that the proposed CNN model achieves stable learning, good convergence, and reliable performance for crack detection in pipeline images.

Figure 4.3 shows Performance Comparison of Crack Detection Systems. The proposed system achieves the highest accuracy of **99.04%** when compared against six existing state-of-the-art pipeline defect detection systems, demonstrating its superior performance for crack detection.

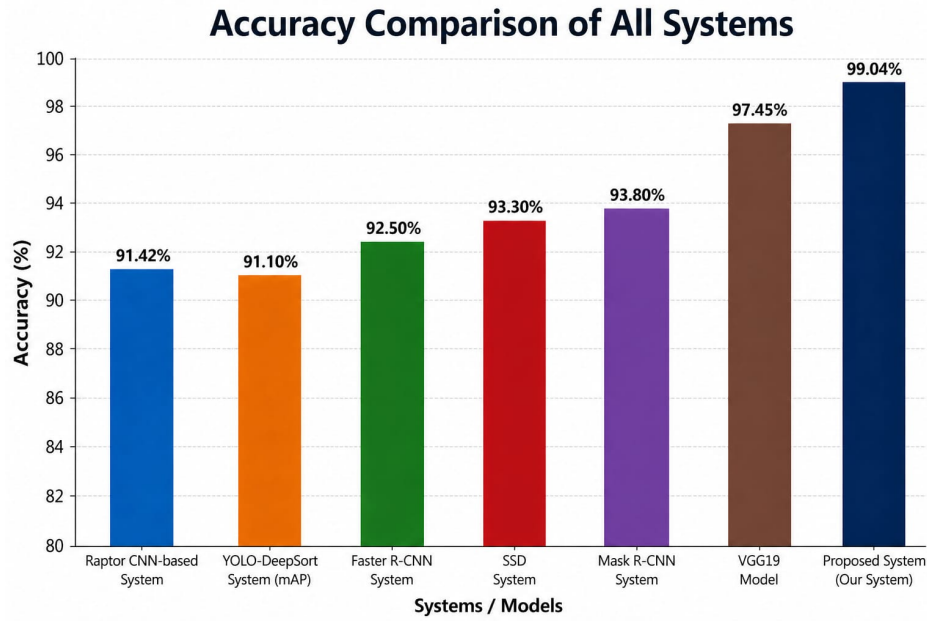


Figure 4.3: performance comparison

The Raptor CNN-based system, a teleoperated robotic platform for drain inspection, achieved a classification accuracy of 91.42% using transfer learning on a custom dataset of drain-blocking materials. The YOLO-DeepSort system, which integrates YOLOv7 with DeepSort for defect tracking, recorded 91.10% mean Average Precision (mAP). The Faster R-CNN system, a two-stage detector that generates region proposals before classification, demonstrated 92.50% accuracy. The SSD system, a one-stage detector balancing speed and accuracy, achieved 93.30% mAP. The Mask R-CNN system, which extends Faster R-CNN with instance segmentation capabilities, attained 93.80% accuracy. The VGG19 model, a deep CNN architecture with 19 layers, achieved 97.45% accuracy for encrustation detection in sewer pipes. The proposed custom CNN model outperforms all these systems by a margin of 1.59% to 7.94%, demonstrating its superior capability for pipeline crack detection while maintaining suitability for edge deployment on resource-constrained hardware such as the Raspberry Pi.

4.3 Integration and System Performance

The integration of detection, navigation, and control modules resulted in a fully functional autonomous inspection robot capable of real-time defect identification and navigation as illustrated in Figure 4.4.

The system demonstrated stable navigation, efficient energy usage, and accurate real-time crack detection, validating its effectiveness in underground pipeline environments. The integration of a custom CNN-based detection model with ultrasonic sensor-based navigation enabled smooth traversal, reliable obstacle avoidance, and minimal human intervention.



Figure 4.4: Robotic system

4.4 User Interface

Figure 4.5 demonstrates the user interface of the pipe inspection robot control panel is designed to provide intuitive and efficient interaction for monitoring and navigation. It features a mode control section that allows the operator to switch seamlessly between automatic mode and manual mode, enabling either autonomous operation or direct user control based on the inspection requirements. The interface also includes an input field to set the pipe diameter, allowing the system to adapt its behavior according to varying pipe sizes for accurate navigation and positioning.

A dedicated live sensor data panel continuously displays real-time measurements, such as front and top distances, providing immediate feedback on the robot's surroundings. This ensures effective obstacle detection and spatial awareness during operation. Additionally, the interface supports manual navigation controls (forward, backward, left, right, and stop), enabling precise movement when manual mode is selected. Visual indicators, such as status messages (e.g., "Outside Pipe - Stopped"), enhance situational awareness and safety. Overall, the UI is structured to deliver real-time monitoring, adaptive configuration, and flexible control, making it suitable for practical pipeline inspection applications.

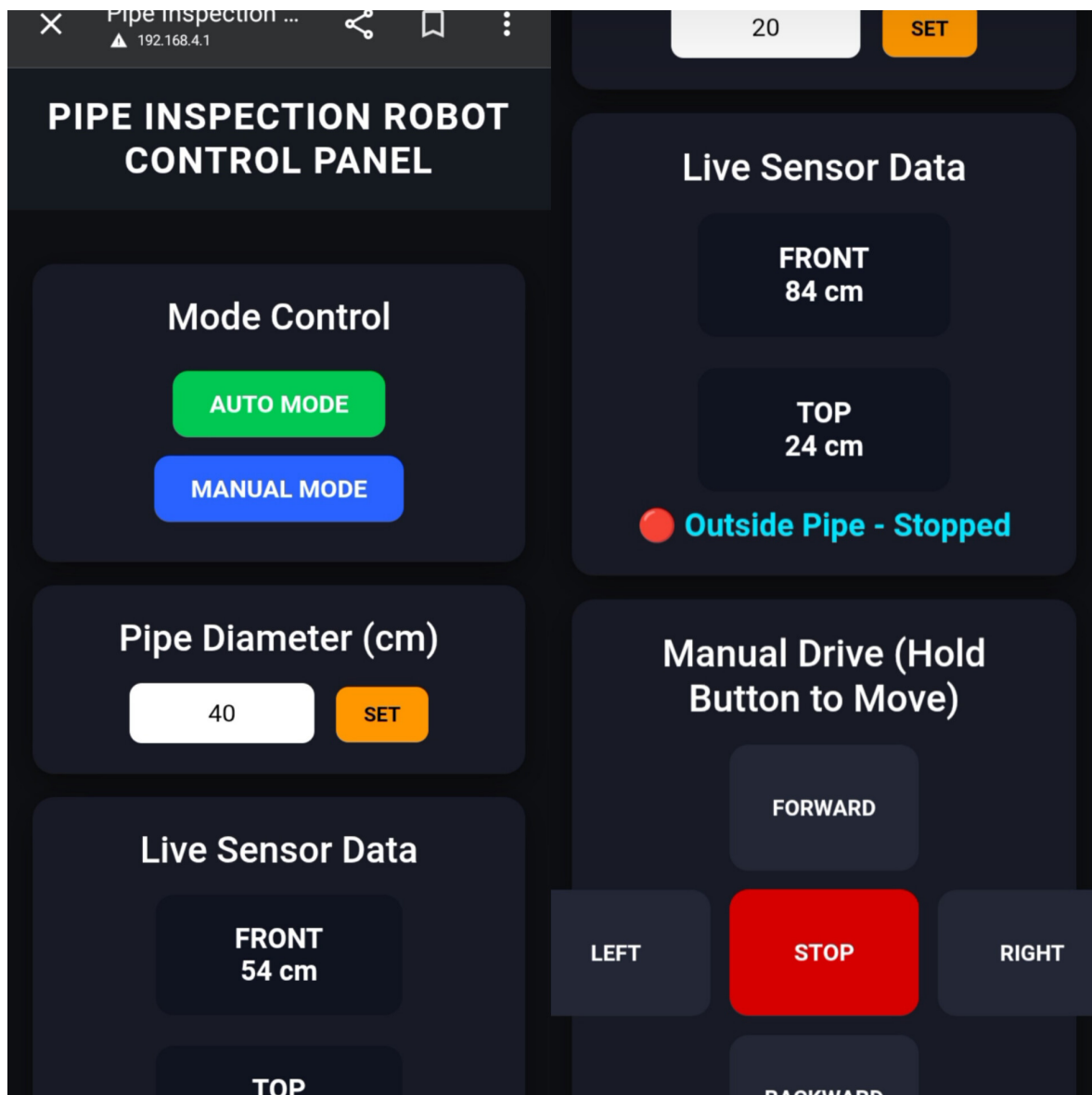


Figure 4.5: operator dashboard

4.5 Discussion

The experimental results confirm that the proposed AI-driven robotic system is a reliable and efficient alternative to manual pipeline inspections. The integration of deep learning-based detection and autonomous navigation enables autonomous, real-time monitoring of pipeline networks with enhanced safety and precision.

Compared to conventional manual inspection methods, the system significantly reduces operational time, improves accuracy, and ensures operator safety in hazardous environments. Furthermore, the modular architecture allows for scalability and potential integration with cloud-based platforms for predictive maintenance and city-scale infrastructure management.

CHAPTER 5

CONCLUSION & FUTURE SCOPE

5.1 Conclusion

The project “Autonomous Robotic System For Pipeline Inspection ” successfully demonstrates the integration of artificial intelligence and embedded robotics for automated pipeline inspection. The developed system utilizes a camera-based robotic platform combined with a deep learning–based crack detection model to perform real-time inspection of pipeline interiors. A Convolutional Neural Network (CNN) deployed on a Raspberry Pi using TensorFlow Lite enables efficient crack detection on edge hardware without the need for cloud processing.

Experimental testing showed that the system is capable of reliably detecting surface cracks under controlled conditions while maintaining stable navigation inside pipelines. The robotic platform, driven by a motor driver–controlled DC motor system, provided smooth movement, and the integration of an ultrasonic sensor ensured obstacle detection and collision avoidance. Wireless communication using an ESP32 module enabled real-time transmission of crack detection alerts and system status for remote monitoring.

The proposed solution effectively addresses key limitations of manual pipeline inspection, including safety risks, limited accessibility, and inconsistent human assessment. By automating the inspection and crack detection process, the system improves reliability, reduces inspection time, and minimizes human involvement in hazardous environments. Overall, the project demonstrates that combining deep learning, computer vision, and robotic automation offers a practical and cost-effective approach for enhancing pipeline maintenance and infrastructure monitoring.

5.2 Future Scope

Although the proposed system exhibits strong performance, several enhancements can further improve its functionality and autonomy. Future developments can focus on the following directions:

- NVIDIA Jetson Nano can be integrated to provide higher processing capability, enabling faster execution of advanced deep learning models such as YOLOv8 for real-time crack detection with bounding box localization.
- A GPS module (NEO-6M) can be added to record the geographic coordinates of detected cracks in outdoor or long-distance pipelines, allowing easier fault localization and maintenance planning.

- The system can be upgraded with a Raspberry Pi Camera v3 (Wide) offering a 120° field of view, which improves visual coverage inside pipelines and enhances defect detection accuracy.
- An IMU sensor (MPU-6050) can be incorporated to monitor robot orientation and tilt, improving navigation stability and providing motion and inclination data during inspection.
- A Li-Ion battery with USB-C charging and an integrated BMS can be used to ensure safer power management, faster charging, and improved portability for field deployment.

With these advancements, the system can evolve into a fully autonomous, intelligent pipeline inspection platform capable of continuous operation, self-diagnosis, and large-scale deployment in smart city infrastructure management.

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